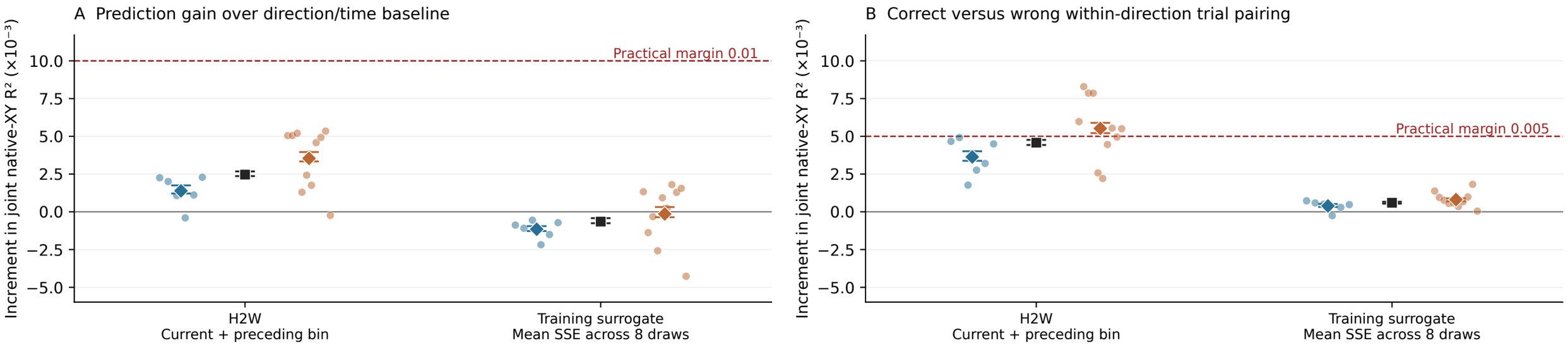
Two-lag history beats training surrogates but trails the blend in prediction

H2W primary gain 0.002465 < 0.010; identity gain 0.004580 < 0.005.

Own training controls share the exact Gram and ridge df (29.127–43.516, 128 direction folds); all selected λ=1.



H2W − blend: primary -0.000214 (both animals and every within-animal omission negative); identity +0.000020.

The small identity gain is fragile in Mihili: mean +0.0000027, fixed-omission range [-0.0000331, +0.0000402].

◆ Mihili mean ◆ Chewie mean ■ Equal-animal mean ○ Individual draw mean

Filled small dots: all 16 sessions. Solid bars: fixed-prediction session-omission ranges. Open dots/dotted spans: 8 draw means and their range.
These ranges are not confidence intervals or p-values. Eight finite controls do not add biological replicates; there are two animals (6 and 10 sessions).
Whole training-trial feature histories are permuted within direction; direction/time structure and correctly paired evaluation trials are retained.
Surrogate correct/wrong-pair SSEs are averaged, not predictions. Own true/null designs share exact capacity; this is not a signal-free or raw-phase null.
H2W uses lag 0 and −1 with the unchanged current-only channel prior. W, four-lag HW and the saved blend remain original, with no reference refit.
Original- and H2W-selected λ coincide for W/HW, so those contrasts overlap exactly. Equal λ or normalized trace does not match capacity across families.
All direct contrasts use complete original support and the same native-XY denominator. Row scales differ; both metrics share each row's full signed scale.
Exposed development data · descriptive model comparisons, not causal lag effects · both practical margins still fail; the saved blend remains the incumbent.